PS02.146

Beyond Best Guesses: A Multi-Level Confidence Assessment Framework for De Novo Peptide Sequencing Results

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De novo sequencing error grammar

De novo peptide sequencing lacks decoy-based FDR control, and existing workarounds don't characterize error types, only overall confidence. Even in high-quality synthetic data, database and de novo search often disagree (Figure 1a). The disagreements aren't random. Levenshtein distance shows a sharp spike at distance two, then a long tail (Figure 1b), hinting most "errors" aren't random mistakes at all. To find out what they actually are, we built peptide similarity alignment (PSA). It contains a Tier definitions based on Levenshtein distance, and event assignment via a Needleman-Wunsch alignment aware of block shuffles and isobaric substitutions. The spike at distance two turns out to be mostly isobaric substitutions and sequence shuffles. These are near-misses, not failures. True multi-event errors fill the long tail. The pattern replicates across datasets, including 100 plant species (Figure 2). This changes what accuracy can mean. PSA resolves amino-acid accuracy by tier and by event, so algorithm developers can see exactly which error type costs how much accuracy (Figure 1e, SWAP/SHUFFLE/ISOBARIC-SUBSTITUTION contributions). Alignment can also be used to show relations. K/R confusion is one of the non-isobaric errors, shifting peptide mass, while other confusions are a methylene sliding along a side chain (V/L, S/T, D/E, A/G, N/Q) (Figure 1f).

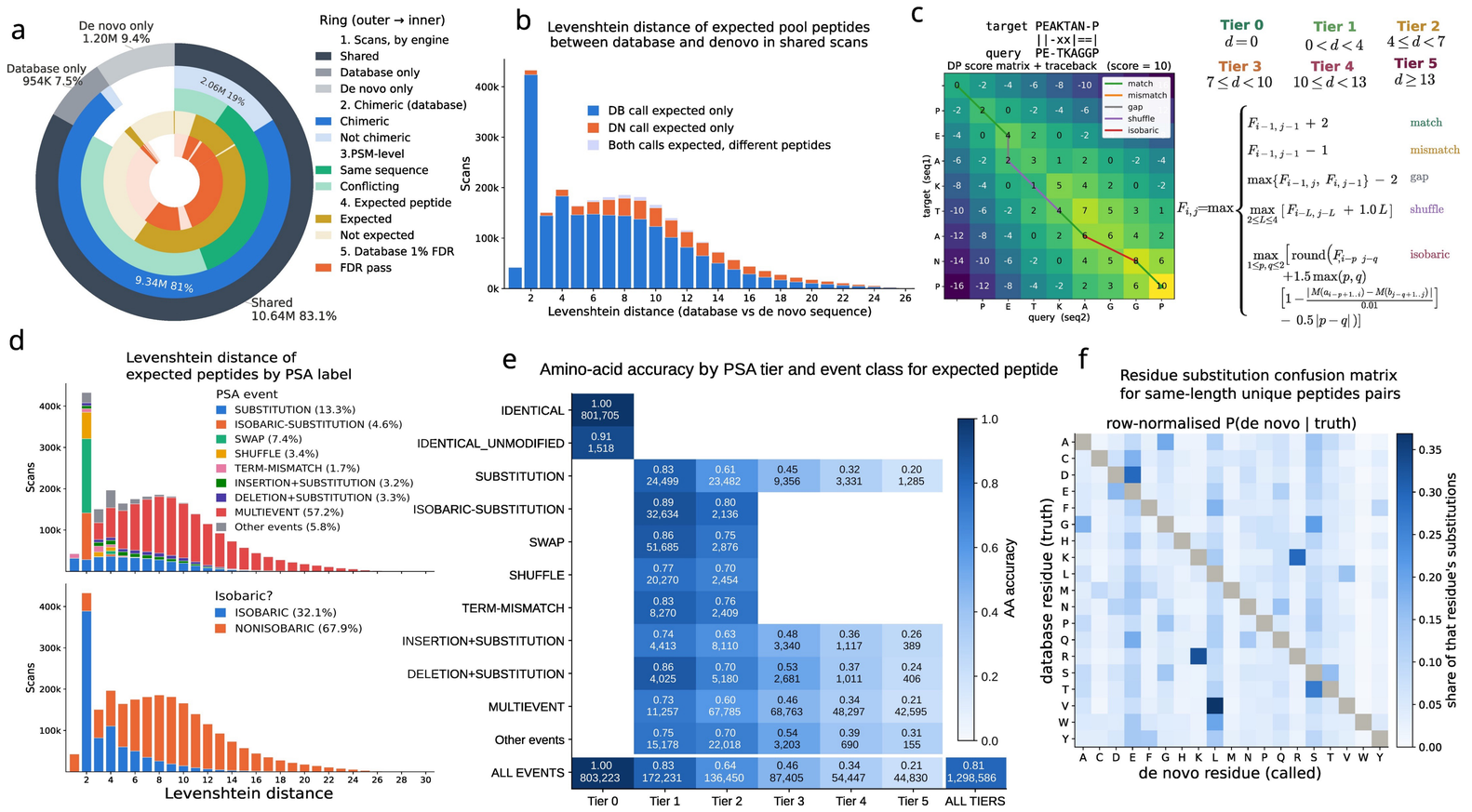


Figure 1. a) The overall difference between the database and de novo results establishes how often the two engines part ways, **b)** The levenshtein distance between the engine calls, restricted to scans with an expected peptide, **c)** The scoring matrix and assignment criteria of peptide similarity alignment, **d)** Relabelling the same distance distribution by peptide similarity alignment event and isobarism, **e)** Amino-acid accuracy split by tier and event turns those labels into a graded statement of correctness. **f)** The substitution confusion matrix between database and de novo residues, restricted to same-length unique peptide pairs. e and f are generated using peptide pairs where database search is 1% FDR controled expected peptides.

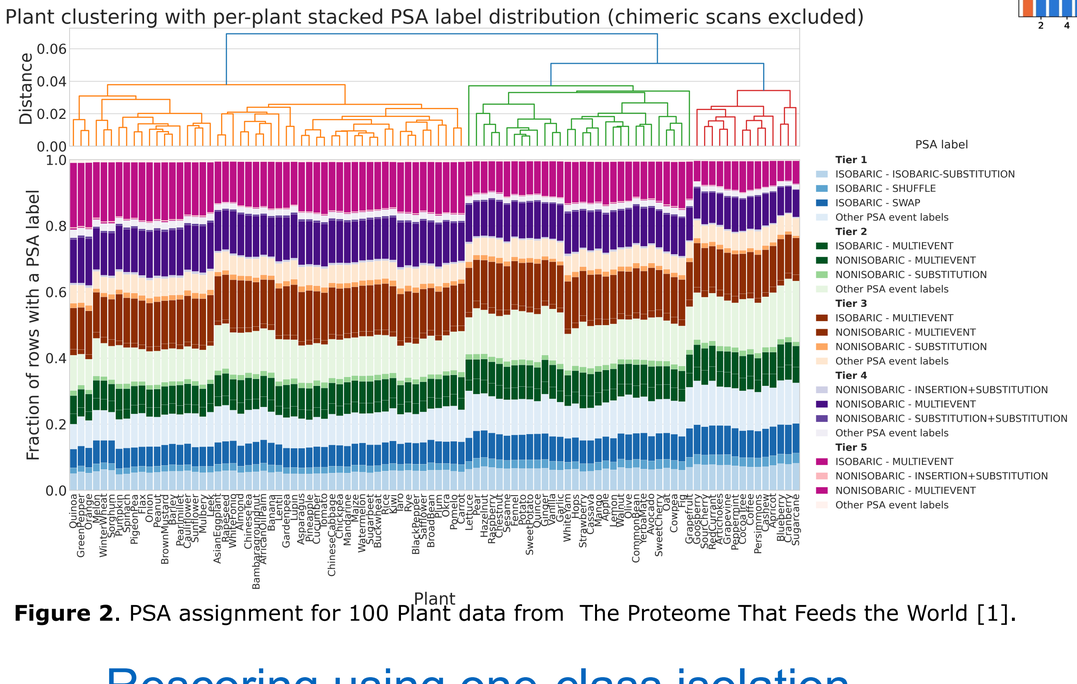


Figure 2. PSA assignment for 100 Plant data from The Proteome That Feeds the World [1].

Rescoring using one-class isolation forest

To understand differences in depth and attempt to separate high quality identification from bad ones, we built a rescoring framework which is similar to classical database search. We used defined 7 features where we observed distinguishable differences for de novo identification comparing the database target-decoy. The features are generated via Oktoberfest. The framework utilizes one-class isolation-forest (IF) (putative true positives only) trained on hypothetical true identifications. A cutoff is defined at the isolation-score value where a two-sample Kolmogorov-Smirnov test finds the largest divergence between the hypothetical true positives and the rest (Figure 3a). Isolation score can capture identifications which has similar true positive like features and makes a bimodal distribution for database and de novo. Moreover, Isolation forest able to separate obviously bad identifications based on the features, yet, it is not sensitive and adds many false positives compared (Figure 3c,d). However, every feature moves monotonically with the score and all populations converge onto the same curve, so the score carries one consistent meaning on both search sides.

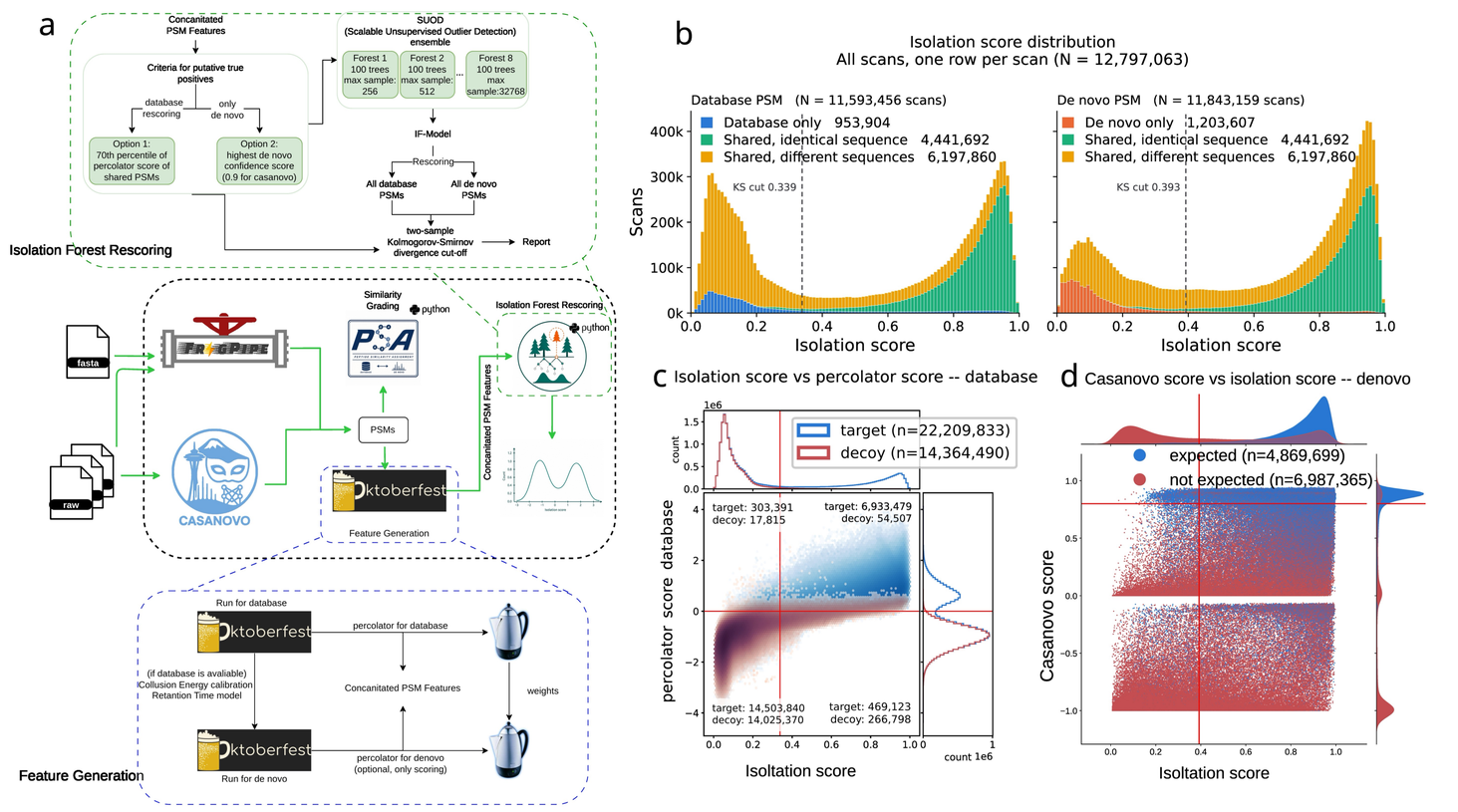


Figure 3 a) The pipeline processes search results through feature generation, similarity grading, and isolation-forest rescoring to a decision. **b)** The isolation-score distributions for the database and de novo sides, broken down by whether the engines agreed, show the trusted population separating from the engine-only ones, with the cutoff derived from the divergence. Only highest score identification kept per duplicated scans. **c)** Percolator score against isolation score on the database side tests with cutoffs. Figures contains chimeric identifications. **d)** Casanovo confidence against the isolation score, with 0.8 cutoff for Casanovo **e)** The isolation forest input features, binned by score with labels.

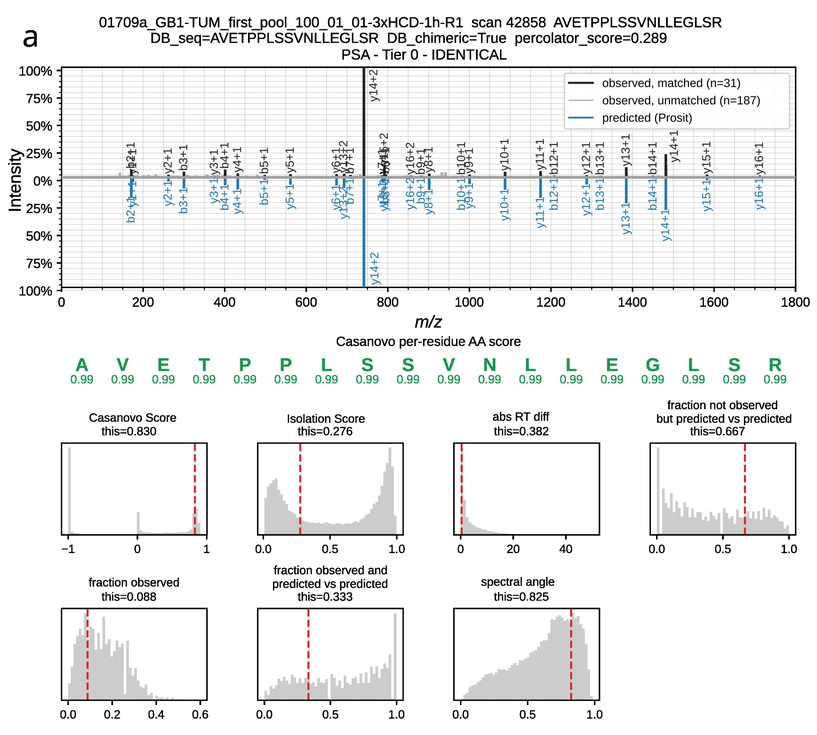


Figure 4 two example mirror plot with features, identification feature is red dashed color

Examples and Limitations

This method does not provide explicit FDR control but filtration based on the identification features. With current approach, one-class IF-base model captures likeliness of given putative true space. However, the biggest limitation of isolation rescore is fundamentally it is not built for the rescoring purposes. It tends to fail on extreme good points (Figure 4a). The other biggest problem is in de novo sequencing it self. For example of Figure b, de novo identifies most of the peptide correctly but fails on four residues, and the mirror spectra still agree very well (PSA Tier 2 MULTIEVENT (SWAP, SUBSTITUTIONS)). With given features, in a classical database target-decoy rescoring this would likely pass FDR. This shows that feature-based de novo rescoring or FDR control is likely to fail here and call a correct, so an orthogonal source of information, or a different approach altogether, is needed to handle these cases.

Conclusion

- PSA grading exposes why errors occur (isobaric swaps, shuffles, multi-events), turning raw mismatches into actionable, tier-specific accuracy metrics.
- One-class IF rescoring separates trustworthy IDs reasonably well but lacks real FDR control and fails on hard cases
- feature-based rescoring alone isn't enough; an orthogonal signal is needed.

Reference:
1. https://www.ls.tum.de/crop-proteomes/the-proteomes-that-feed-the-world/